

Guidelines for Undergraduate B.Tech. CSE (Data Science) Final Year Thesis Formatting

Department of Computer Science and Engineering(Data Science)

December 2025

1 Introduction

These guidelines provide a standardized format for the final year thesis of B.Tech. CSE (Data Science) students. The thesis must demonstrate original work in data science applications, such as machine learning, big data analytics, or AI-driven systems. Adherence to these rules ensures clarity, professionalism, and academic integrity. The document should be approximately 60–80 pages, excluding appendices, declaration, Certificate, Table of Contents pages.

2 Overall Formatting Requirements

- **Paper Size and Margins:** A4 paper with 1-inch margins on all sides (top, bottom, left, right).
- **Font:** Times New Roman, 12pt for body text; 14pt bold for chapter titles; 12pt bold for section headings.
- **Line Spacing:** Strictly 1.5 line spacing throughout the document, including abstract and references. Use single spacing for captions, footnotes, and long quotations.

- **Alignment:** Justified text with 0.5-inch indentation for new paragraphs (no extra space between paragraphs).
- **Page Numbering:** Roman numerals (i, ii, ...) for front matter (title page to abstract); Arabic numerals (1, 2, ...) starting from Chapter 1.
- **File Format:** Submit as PDF compiled from LaTeX.

3 Thesis Structure

The thesis must follow this structure:

1. Front Matter:

- Title Page: Include title, student name, USN, supervisor name, department, institution, and year.
- Declaration: Student declaration of originality.
- Certificate: Supervisor's certificate of approval.
- Acknowledgements: Thank contributors (optional, 1 page max).
- Abstract: 300–500 words summarizing problem, methods, results, and conclusions.
- Keywords: 5–7 relevant terms.
- Table of Contents: List all chapters, sections, and page numbers.
- List of Figures, Tables, Abbreviations.

2. Main Body:

- Chapter 1: Introduction (Background, Objectives, Problem Statement, Summary).
- Chapter 2: Literature Review (Related Work, ML in Domain, Limitations).
- Chapter 3: Methodology (System Architecture, Project Planning, Requirements).

- Chapter 4: Implementation (Algorithm, Integration, Testing).
- Chapter 5: Results and Discussion (Experimental Results, Evaluation, Validation, Comparative Analysis, Optimization).
- Chapter 6: Conclusion and Future Work (Conclusion, Key Learnings, Future Work, SDGs).

3. Back Matter:

- References: IEEE style, using BibLaTeX.
- Appendices: Code samples, surveys, detailed matrices.

4 Equations, Tables, Figures, and Algorithms

- **Equations:** Use `amsmath` package. All equations must be centered and numbered chapter-wise (e.g., (1.1)). Refer in text as “Equation 1”.

- Example:

$$E = mc^2 \tag{1}$$

As shown in Equation 1.

- **Tables:** Captions above the table, numbered chapter-wise (e.g., Table 1.1). Use `tabular` environment. Refer as “Table 1”.

- Example:

Table 1: Sample Table

Item	Value
A	1
B	2

See Table 1.

- **Figures:** Captions below the figure, numbered chapter-wise (e.g., Figure 1.1). Use `graphicx`. Refer as “Figure 1”.
- Example:

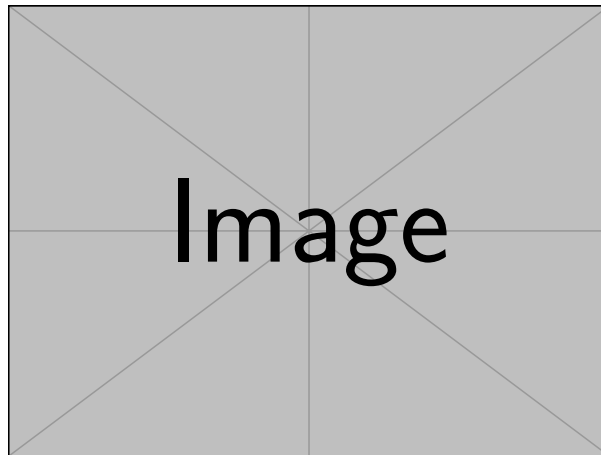


Figure 1: Sample Figure

As illustrated in Figure 1.

- **Algorithms:** Use `algorithm` and `algpseudocode` packages. Numbered chapter-wise (e.g., Algorithm 4.1). Refer as “Algorithm 1”.
- Example:

Algorithm 1 Sample Algorithm

Input: Data

Output: Result

Refer to Algorithm 1.

5 References and Citations

Use IEEE citation style. Cite in text as **example**. All sources must be listed in the References section. Minimum 30–50 references, prioritizing recent peer-reviewed articles from standard publishers like *IEEE*, *Springer*, *Elsevier*.

6 Additional Guidelines

- **Lists:** Use `itemize` for bullets, `enumerate` for numbers. Nested lists allowed.
- **Code:** In appendices, use `listings` package with language highlighting.
- **Plagiarism:** Must be below 10%; include report.
- **AI Plagiarism:** Must be below 10%; include report.
- **Length per Chapter:** Introduction (4–6 pages), Literature (8–12), etc.
- **Submission:** Hard copy (3) and soft copy on USB(L^AT_EX source files and PDF).